Name:________________________________

1 2D Plotting - Numerical Derivative Errors

There are problems where you have to compute the derivative of a given signal, but you don't know its underlying function. Let $f(x) = \sin(2\pi x)$. Let's approximate its derivative and compute the error between the approximation and the actual derivative by doing the following:

1. Define a uniform grid $\{x_i\}_{i=1}^{1000}$ over $[0, 5]$ with 1000 points. (i.e: `x = linspace(0,5,1000)`).
2. Evaluate $f(x)$ in a variable `f` (i.e. `f = sin(2 * pi * x)`), and evaluate the actual derivative $f'(x)$ in a variable `df_actual` (i.e. `df_actual = 2 * pi * cos(2 * pi * x)`)
3. Set `h` to the distance between grid points $x_2 - x_1$ (i.e. `h = x(2) - x(1)`)
4. Use the forward difference formula to approximate the derivative $f'(x)$, store the result in `df_approx`

$$f'(x_i) \approx \frac{1}{h}(f(x_{i+1}) - f(x_i)) \tag{1}$$

   (i.e. `df_approx = (f(2:end) - f(1:end-1)) / h;`)

2 Questions

1. (8 pts) Write **MATLAB code** for a plot with two subplots in two rows (**include the title and legends**):
   (a) The first subplot, plot `df_actual(2:end)` in blue and line width of 1, as well as `df_approx` in red and line width of 1. Limit the y-axis to $[-8, 8]$.
   (b) The second subplot (below the first) plot `abs(df_actual(2:end) - df_approx)`, the error between the numerical and exact derivatives. Limit the y-axis to [0, 0.2]

2. (2 pts) What is the maximum error in `abs(df_actual(2:end) - df_approx)` (give answer up to 4 decimal places)? (Hint: use `max(.)`)